

Patrick Cherry, PhD

✉ patrick [at] pcherry [dot] org | 📞 upon request | 🏢 Principal Scientist | Oncology | 🏠 BillionToOne | 📍 San Francisco, California |
🌐 pcherry.io | 🐙 github.com/pdcherry | in linkedin.com/in/p-cherry | 🇺🇸 United States Citizen | Updated on 2026-10-05

PhD data scientist and professional tool-builder and science communicator. I bring the incisive critical thinking of an -omics-trained scientist to data science. Skilled at data visualization, pipeline building, and custom tool building and validation. I am dedicated to serving as the bridge between complex data problems and action-oriented leadership seeking data-driven strategy.

🏢 Industry Experience

- Current
|
Oct 2026
- **Principal Scientist, Oncology**
BillionToOne, Inc. 📍 Menlo Park, California
 - **Senior Scientist, Oncology**
BillionToOne, Inc. 📍 Menlo Park, California
 - Tech lead and program manager for multi-quarter cfDNA genomics assay expansion (Select v3); led the launch and CAP/CLIA validation of Northstar Select Pharmacogenomics (PGx) in under two months
 - Implemented department-wide copy number database using extract-transform-load (ETL) procedure on production data
 - Ideated, coded, and validated [patented](#) method for tumor fraction estimation from plasma aneuploidy data using Gaussian mixture models, custom model selection metric, and custom consensus logic
 - Answered business and product design questions from leadership about un-met customer need with internal reports on Notion written in reproducible coding notebooks using Python and R in a HIPAA-compliant environment
 - As hiring manager, recruited and hired four Research Associates; managed the Statistics interview
- Apr 2024
|
Dec 2021
- **Senior Scientist, Genomics**
Twist Bioscience, Inc. 📍 South San Francisco, California
 - Invented and introduced multiple new products to market, yielding \$1M+ in new revenue as *Tech Lead*, including: *Pan-cancer cfDNA* (v1 & v2) ([patented](#)), *CNV Controls*, *RNA Fusion Controls*, RNA-seq, and [patented adapter removal](#)
 - Answered feasibility questions with [Design of Experiment](#) principles and data analysis using R and Python.
 - Wrote, documented, and maintained production QC pipelines for sub-components and finished good QC; trained end-users in Manufacturing and Quality on their use, interpretation, and reporting.
 - Maintained production code ([git](#) version control with code review and team best practices) with CD (Github actions); operated in AWS & Databricks high-performance compute environment using Unix shell
 - Answered manufacturing & business questions using SQL / Snowflake database of production events and customer data
- Jan 2021
|
Apr 2019
- **Scientist I, Next Generation Sequencing (NGS)**
Zymergen, Inc. 📍 Emeryville, California
 - Boosted pan-project NGS genotyping success by 45% using [Design of Experiment](#) and Estimated Marginal Means modeling for causal inference; Guided demanding and diverse internal customers on complex NGS experiments
 - Built and disseminated reproducible statistical templates in [Rmarkdown](#) and [JMP](#) for self-serve NGS data exploration
 - Rendered strain build and plasmid build reports from LIMS SQL database in [MariaDB](#) and R / [ggplot2](#)
 - Designed, built, and demonstrated a [patented](#) counterselectable marker for the non-model microbe *Bacillus subtilis*

🎓 Education

- Apr 2019
|
Aug 2013
- **PhD in Molecular Biology**
University of Colorado School of Medicine 📍 Aurora/Denver, Colorado
 - Ph.D. in Molecular Biology. Advisor: Jay Hesselberth, PhD. Key courses: 6606 Statistics; 7621 Genome Analysis Workshop.
 - Thesis: RNA terminus chemistry potentiates decay events targeting *HAC1* mRNA during the Unfolded Protein Response
- May 2013
|
Aug 2009
- **BA in Molecular Biology with Distinction, minor in Mathematics**
Hendrix College 📍 Conway, Arkansas
 - Senior Capstone Project showing changes of key stress response gene expression to oxidative stress in liver cells
 - Key courses: Probability & Statistics; Discrete Mathematics; Adv Biological Chemistry, Analytical Chemistry